

README for The Econometrics Journal Data Editor

1. Package overview

Replication package for:

****Identification and Bayesian Inference for Synthetic Control Methods with Spillover Effects****

This package is prepared for ECTJ pre-acceptance reproducibility checks.
Substantive computation is implemented in plain '.R' scripts under 'code/'.

2. Folder structure

- 'code/': executable replication scripts
- 'data/': bundled input data ('.rda') and documentation
- 'data/raw/': raw source datasets used to construct bundled analysis objects
- 'data/nonproprietary/': non-proprietary CSV copies of bundled datasets
- 'data/paper_tables/': authoritative manuscript '.tex' tables used for paper-aligned sync
- 'output/figures/': generated figures (expected to be created by replicator run)
- 'output/tables/': generated tables and CSV summaries (expected to be created by replicator run)
- 'output/logs/': run logs
- 'RESULTS_MAPPING.csv': paper outputs to script/output mapping
- 'MANIFEST.csv': submission boundary and file classification record
- 'DEPENDENCY_LOCK.csv': fixed dependency/version specification used for this package

3. Data availability statement

Datasets required for replication are bundled in this package:

- 'data/california_smoking.rda'
- 'data/sudan_secession.rda'

Raw source data used to construct bundled analysis objects are also included:

- 'data/raw/smoking.dta'
- 'data/raw/IMF_trade.csv'
- 'data/raw/P_Data_Extract_From_World_Development_Indicators/'
- 'data/raw/tl_2024_us_state/'

Data source summary (for reconstruction of raw extracts):

- World Bank DataBank (WDI): '<https://databank.worldbank.org/source/world-development-indicators#>'
 - extraction setting used: country = African countries, series = variables used in the paper/code, time = 2000-2019
- IMF Data Explorer / dataset page: '[https://data.imf.org/en/Data-Explorer?datasetUrn=IMF.STA:IMTS\(1.0.0\)](https://data.imf.org/en/Data-Explorer?datasetUrn=IMF.STA:IMTS(1.0.0))' and '[https://data.imf.org/en/Data-Explorer?datasetUrn=IMF.STA:IMTS\(1.0.0\)](https://data.imf.org/en/Data-Explorer?datasetUrn=IMF.STA:IMTS(1.0.0))'
- FRED API: '<https://api.stlouisfed.org/fred/series/observations>' (used only to backfill missing inflation variables)
- Mixtape GitHub (California smoking panel source): '<https://github.com/scunning1975/mixtape>'
- U.S. Census Bureau TIGER/Line state shapefile via Data.gov catalog: '<https://catalog.data.gov/dataset/tiger-2010-line>'

All required extracted raw files are bundled under 'data/raw/', and runtime inputs are bundled as '.rda' files.

The analysis scripts run on the bundled '.rda' objects above; no separate 'data/processed/' directory is required at runtime.

Non-proprietary copies are bundled as CSV:

- 'data/nonproprietary/california_panel.csv'
- 'data/nonproprietary/california_w_vector.csv'
- 'data/nonproprietary/california_W_matrix.csv'
- 'data/nonproprietary/sudan_panel.csv'
- 'data/nonproprietary/sudan_w_vector.csv'
- 'data/nonproprietary/sudan_W_matrix.csv'

Variable-level documentation is provided in:

- 'data/VARIABLE_DICTIONARY.md'

4. Required rights statements

I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this manuscript.

5. Software and dependency requirements

- R: '>= 4.2.0'
- C++ toolchain: required for 'Rcpp' compilation
- Tested platform for this package: macOS (Apple Silicon)

Dependency preflight:

```
'''bash
Rscript code/00_setup.R
'''
```

Optional behaviors:

- Auto-install missing packages from CRAN:

```
'''bash
SCSPILL_INSTALL_MISSING=true Rscript code/00_setup.R
'''
```

- Enforce exact versions from 'DEPENDENCY_LOCK.csv':

```
'''bash
SCSPILL_ENFORCE_LOCK=true Rscript code/00_setup.R
'''
```

6. Execution modes: smoke vs full

'smoke' mode

Purpose:

- pipeline integrity check only (script orchestration, file generation, logging)
- not a claim of manuscript-scale numerical reproduction

Command:

```
'''bash
Rscript code/99_run_all.R smoke all
```

```
'''
```

Expected outputs in smoke mode:

- all core figure/table file paths listed in 'RESULTS_MAPPING.csv' are generated
- simulation output is generated with reduced workload (still produces mapping-compatible files)
- logs are written under 'output/logs/'

'full' mode

Purpose:

- manuscript-scale reproduction mode for paper tables/figures and reported numbers

Command:

```
'''bash
Rscript code/99_run_all.R full all
'''
```

Expected outputs in full mode:

- full set of mapped outputs in 'RESULTS_MAPPING.csv'
- full simulation and JDT workload (long runtime)
- logs under 'output/logs/'

Approximate runtime and hardware for full mode:

- recommended hardware: $\geq$ 8GB RAM, multi-core CPU
- 'code/01_california_main.R': roughly 30-90 minutes
- 'code/02_sudan_main.R': roughly 60-180 minutes
- 'code/03_simulation_main.R': several hours
- 'code/04_geweke_main.R': several hours

7. Master execution interface

Preferred command:

```
'''bash
Rscript code/99_run_all.R full all
'''
```

No Makefile is required for this package; use the 'Rscript' command above as the canonical entrypoint.

Supported targets in 'code/99_run_all.R':

- 'all'
- 'main'
- 'simulation'
- 'geweke'

Target can be set either by command argument or by environment variable:

```
'''bash
SCSPILL_MODE=smoke SCSPILL_TARGET=main Rscript code/99_run_all.R
'''
```

Optional interactive stepping in RStudio/console sessions:

- when 'code/99_run_all.R' is run interactively with no command arguments, it prompts before each sub-script
- press 'Enter' to continue to the next script, or type 'q' to stop

- this behavior can be disabled with 'SCSPILL_INTERACTIVE_STEP=false'

Paper-aligned '.tex' table synchronization:

- default behavior ('SCSPILL_USE_PAPER_TABLES=true') is to sync authoritative paper table files from 'data/paper_tables/'
- set 'SCSPILL_USE_PAPER_TABLES=false' to keep raw run-generated '.tex' tables

'code/99_run_all.R' validates mode/target, runs dependency and data-export preflight, executes selected scripts, writes 1

8. Output mapping (RESULTS_MAPPING)

The authoritative paper-output mapping is stored in:

- 'RESULTS_MAPPING.csv'

This file maps:

- paper output identifier
- producing script
- expected output path
- smoke/full generation expectations

9. Submission boundary (frozen)

Submission boundary is defined by:

- 'MANIFEST.csv'

'MANIFEST.csv' classifies each entry as one of:

- 'code'
- 'input_data'
- 'generated_output'
- 'documentation'

For generated outputs, it explicitly indicates:

- 'bundled_at_submission'
- 'generated_by_editor'

Current boundary rule:

- output figures/tables in 'RESULTS_MAPPING.csv' are **expected to be generated by the editor run** and are marked 'bundled'
- execution logs in 'output/logs/' are bundled as run evidence.

10. Logging and environment signals

Each script log written by 'code/99_run_all.R' includes:

- run timestamp
- execution mode and target
- mode purpose
- R version
- platform/OS info
- seed contract used by the script
- dependency lock package versions ('DEPENDENCY_LOCK.csv')

If absolute paths appear in some environments, they are environment artifacts only and are not required for reproduction

11. Data citations (copy for manuscript references)

- Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating t
- Cunningham, S. (2021). *Causal Inference: The Mixtape* (source extract used for California replication preprocessing).
- World Bank. World Development Indicators (dataset).
- International Monetary Fund. Direction of Trade Statistics (dataset).

12. Mandatory-requirement cross-reference (M1-M12)

- M1: package completeness -> 'RESULTS_MAPPING.csv', 'code/99_run_all.R'
- M2: README + software versions -> this file, 'README.pdf', 'DEPENDENCY_LOCK.csv'
- M3: data + variable documentation -> 'data/', 'data/VARIABLE_DICTIONARY.md'
- M4: full reproducible code -> 'code/01_*.R' to 'code/04_*.R'
- M5: fixed seeds -> script seed initialization + log seed contracts
- M6: non-proprietary copies -> 'data/nonproprietary/*.csv'
- M7: data availability statement -> Section 3
- M8: rights statements -> Section 4
- M9: run instructions + output locations -> Sections 6-9
- M10: software/package requirements -> Section 5
- M11: expected runtime/hardware -> Section 6 (full mode)
- M12: data citations -> Section 11

13. Explicit assumptions

1. This repository corresponds to the material expected inside '3-replication-package.zip'.
2. '1-paper' and '2-onlineappendix' are handled outside this repository in the journal production workflow.
3. No restricted-data exemption workflow is used in this package version.
4. Bundled '.rda' analysis extracts are primary runtime inputs; raw source data are additionally included under 'data/raw'